Prediction Is Not Fact

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected prediction-checking record.

Identify the two quantities, test equality, preserve unresolved observations and audit a rewritten score.

Keep one habit: Keep prediction, observation evidence and comparison status in separate fields.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Boundaries and 'Not Sure'

Build a Human Classifier

Repeating Efficiently

W1 · Separate input, prediction and observation

> Use the supplied Ramp1 reference and keep the two measured quantities distinct.

Your evidence record

- Ramp1 reference: allowed ramp heights are whole centimetres 1–10; predict far at height 6 cm or more and near otherwise. The separate observation policy is far at travel distance 4 floor units or more, near otherwise. Worked records A=(height 4 cm,distance 3 units), B=(6 cm,3 units), C=(8 cm,5 units). Keep each input and later observation linked by its trial ID.

- For Ramp1 trial B, height is 6 cm and the later travel distance is 3 units. State the **input**, **prediction**, **observation**, observed category and comparison status.

  Explain why the **prediction** rule and **outcome rule** use different measurements.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Check the observation boundary

> Apply the stated inclusive travel-distance rule.

Your evidence record

- Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the **outcome rule**. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Preserve missing and conflicting evidence

> Request the original trial record before judging the prediction.

Your evidence record

- D predicts near but has no distance record. E predicts far but its two distance records say 3 and 4 units.

  State each comparison status, request useful **evidence**, and explain why neither guessing nor a new trial can establish the old trial's outcome.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Audit a rewritten prediction record

> Retain the original model output and report only established comparisons.

Your evidence record

- After seeing B's observed near category, a learner replaces Ramp1's far **prediction** with near. They call the new entry a corrected **observation** and report a perfect result for A through C.

  Explain the two record-keeping errors and write a defensible replacement report.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the check to drying predictions

> Keep one first attempt before hints or feedback.

Read and explain independently

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the check to drying predictions

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Check a boundary without changing the rule for **Prediction**s and **observation**s.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Prediction**s and **observation**s. Proposed explanation: “Running the rule correctly proves the **prediction** correct”. Decide whether this explanation is supported.

Point to the step or supplied **fact** that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A weather helper predicts tomorrow's rain.

Is this a **fact** about tomorrow? How can it be checked?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A model predicts rain tomorrow. Tomorrow is dry. Was the **prediction** an **observation** of tomorrow? Explain what new **evidence** settles the result.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 17 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

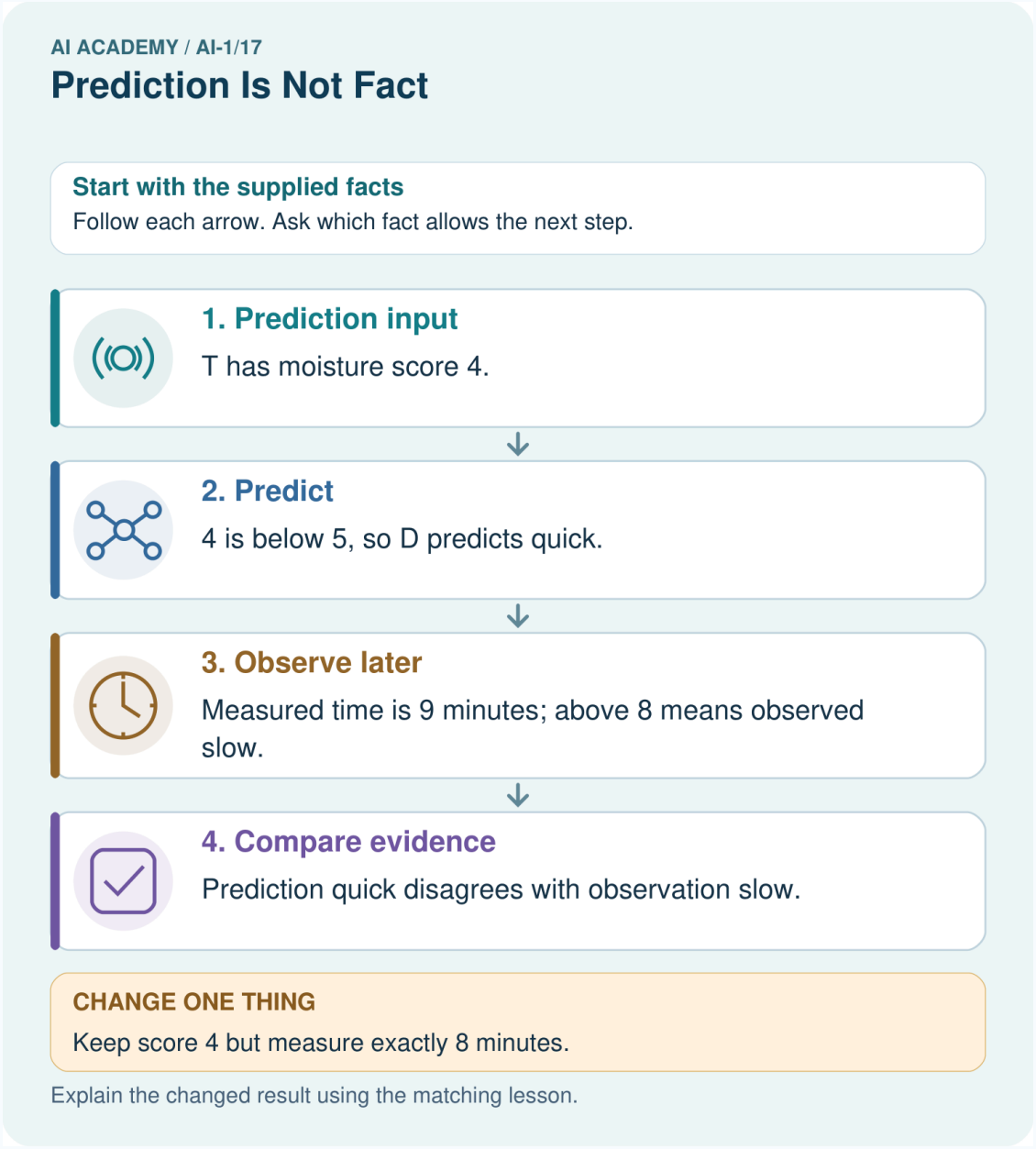
> Keep the explanation and the support you used.

Use this when helpful

A fictional drying trial uses a fixed rule and later measures what actually happened.

Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which field changes the **observation** label, and which field changes the model **prediction**?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional drying trial uses a fixed rule and later measures what actually happened.
- Which labelled step matches this detail? **Prediction** quick disagrees with **observation** slow.
- 1. Predict
- 2. Compare **evidence**
- 3. Observe later
- 4. **Prediction input**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare **evidence Prediction** quick disagrees with **observation** slow. The moisture score generates a **prediction**; it is not the observed drying time.

The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units. Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Changed-case question

- Changed-case practice: Keep score 4 but measure exactly 8 minutes. Which conclusion is supported by the supplied **fact**s?
- 1. The observed label becomes quick and agrees with the **prediction**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those **fact**s.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The observed label becomes quick and agrees with the **prediction**. Above 8 is strict: equality is quick under this policy.

Repair and save your work

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

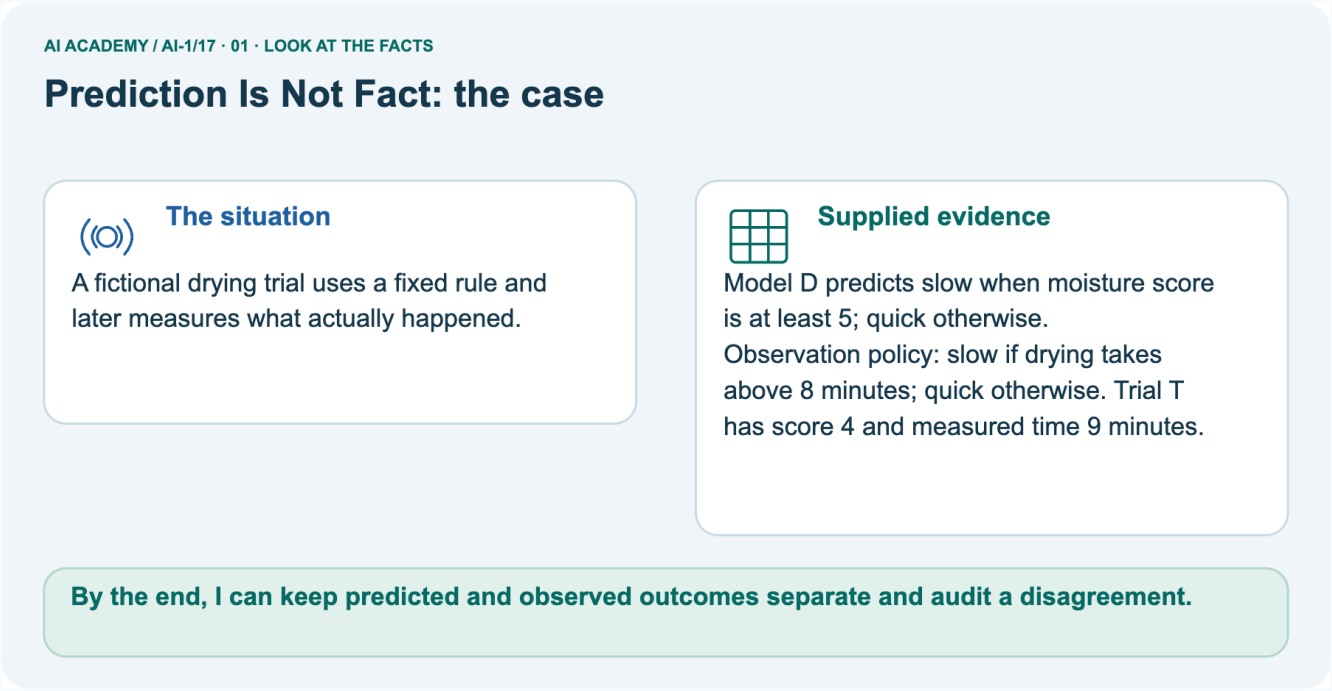
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Prediction Is Not Fact: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

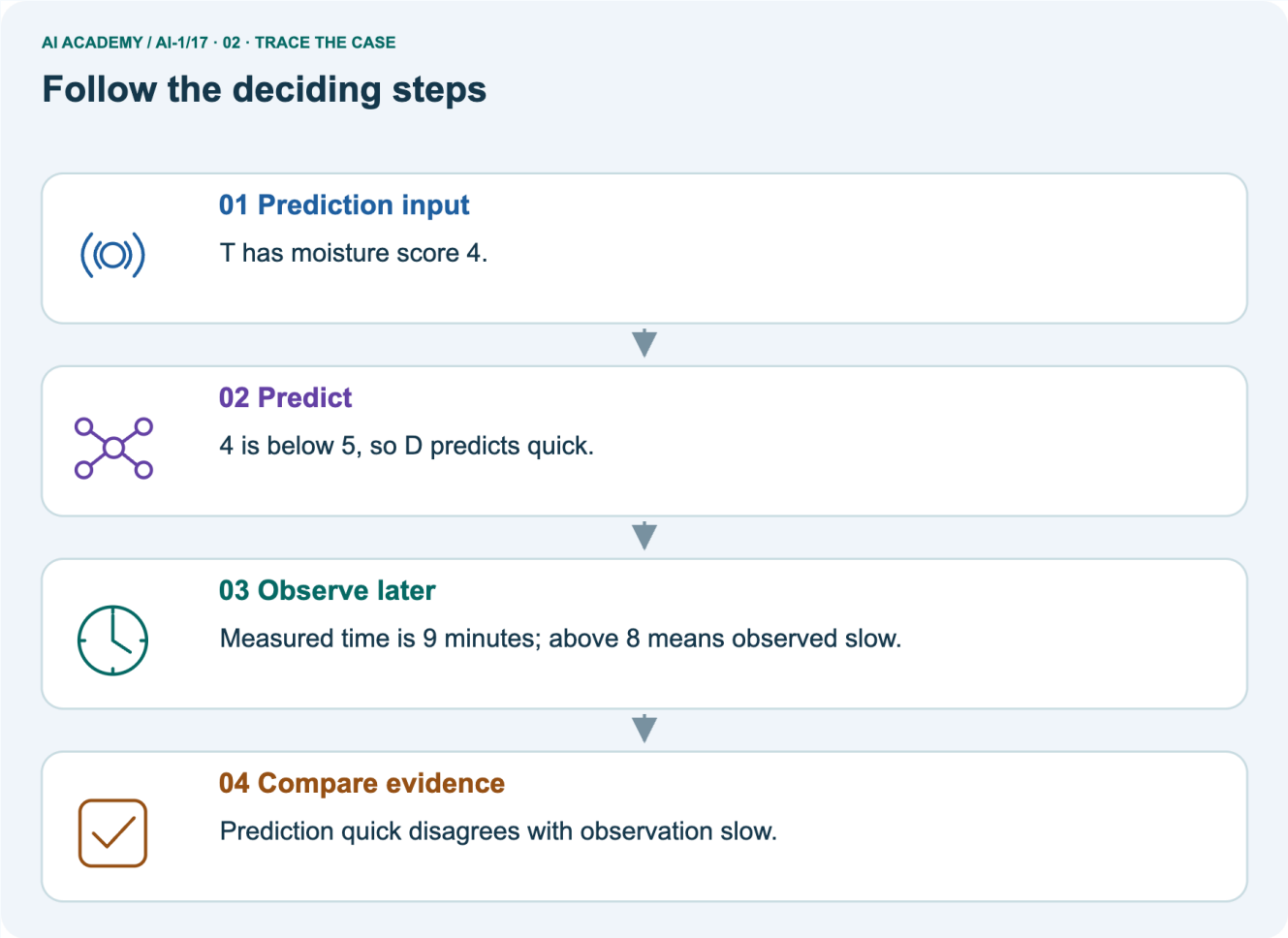
A fictional drying trial uses a fixed rule and later measures what actually happened. Supplied **fact**s: Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

Point to the given **fact** that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Prediction input: T has moisture score 4. → Predict: 4 is below 5, so D predicts quick. → Observe later: Measured time is 9 minutes; above 8 means observed slow. → Compare **evidence**: **Prediction** quick disagrees with **observation** slow.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two records meet at a comparison

AI ACADEMY / AI-1/17 · 03 · INSPECT THE LESSON MODEL

Two records meet at a comparison

Before the trial	After the trial	Comparison
A: height 4 cm	Distance 3 units	Near matches near
B: height 6 cm	Distance 3 units	Far differs from near
C: height 8 cm	Distance 5 units	Far matches far

Ramp1 predicts from height: A near, B far, C far. The observation is travel distance. Apply the distance outcome rule separately, then compare categories for the same trial.

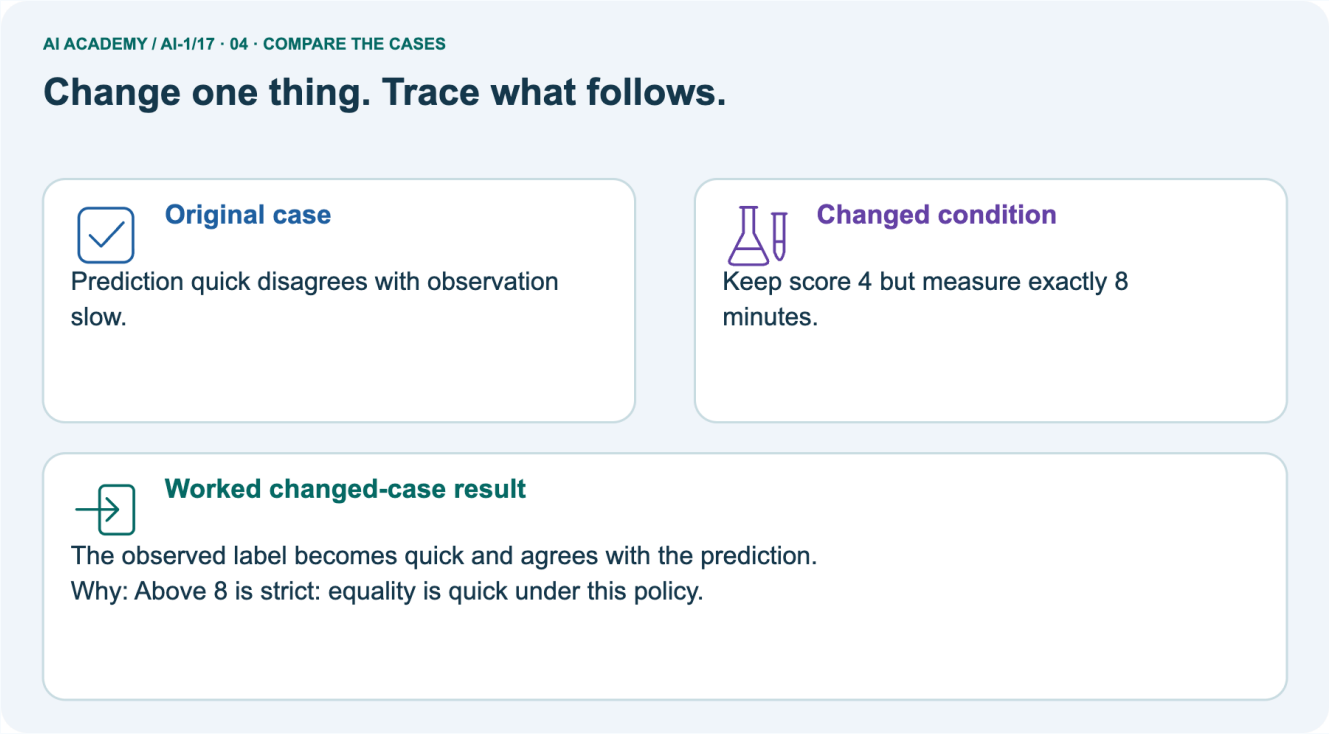
Follow the labelled picture with the matching explanation. Open the vector diagram

Two records meet at a comparison. A: height 4 cm | Distance 3 units | Near matches near. B: height 6 cm | Distance 3 units | Far differs from near. C: height 8 cm | Distance 5 units | Far matches far

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

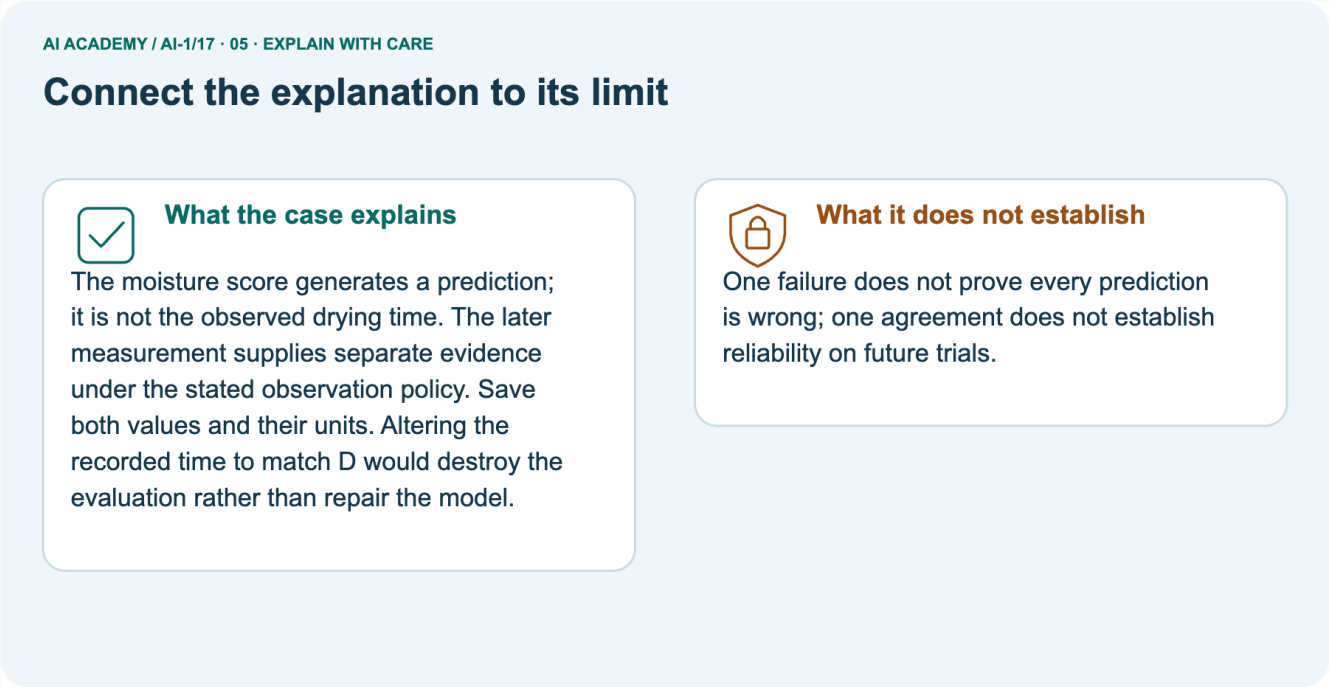
Original case: **Prediction** quick disagrees with **observation** slow. Changed condition: Keep score 4 but measure exactly 8 minutes. Expected result: The observed label becomes quick and agrees with the **prediction**.

Reason: Above 8 is strict: equality is quick under this policy.

Predict before checking the worked changed-case result. Name what changed and what the supplied **fact**s allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

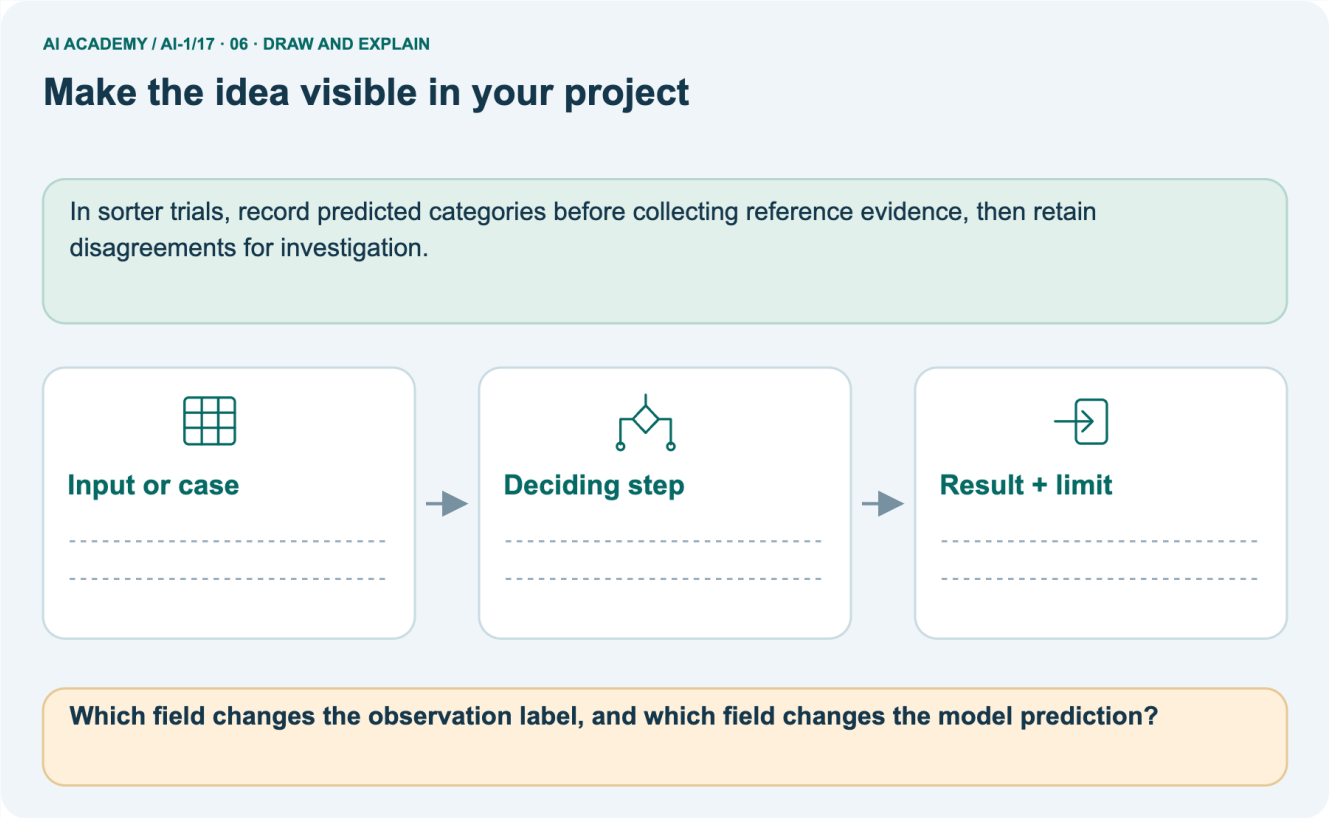
The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model. Limit: One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation. Draw your **input**, deciding step, result and limit.

Supported question: Which field changes the **observation** label, and which field changes the model **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Prediction Is Not Fact: the case

02. Follow the deciding steps

03. Two records meet at a comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.